

Aniruddhan Ganesaraman

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Education

Ph.D., Statistics and Operations Research (STOR) Dept. of STOR, The University of North Carolina at Chapel Hill Advisors: Prof. Patrick Lopatto & Prof. Michael Kosorok	Aug 2024 – May 2029 Chapel Hill, NC
M.S., Statistics, Analytics, and Data Science Dept. of STOR, The University of North Carolina at Chapel Hill Advisor: Prof. Vidyadhar Kulkarni Thesis: <i>Data-Driven Block Replacement Scheduling</i> (defended Mar 2026; arXiv)	Aug 2024 – May 2026 Chapel Hill, NC
M.Stat. (Master of Statistics) , Advanced Probability Theory Specialization Indian Statistical Institute Advisor: Prof. Arijit Chakrabarty Thesis: <i>Clustering of Large Deviation Events</i>	Aug 2022 – May 2024 Kolkata, India
B.Sc. (Hons.), Mathematics and Computer Science Chennai Mathematical Institute	Aug 2019 – May 2022 Chennai, India

Summary

PhD student in STOR working on **causal inference**, **adaptive policy learning**, and **stochastic modeling & simulation**, driven by finding solutions to problems with meaningful real-world impact, from hospital operations to public health. Experienced communicating technical work to varied audiences through teaching, conference talks, and UNC's Three Minute Thesis finals.

Papers & Preprints

Discussion of “Hypothesis Testing for Treatment Effect Heterogeneity via an Adaptively Augmented Debiased Estimator” ASA Discoveries . Joint work with Annika Cleven , Xinyi Li , Margaret Hoch , Keshav Saxena , Matthew Egberg , Bryce Rowland , and Michael R. Kosorok .	Accepted Aug 2026
Heterogeneous Causal Effects of Social Determinants of Health on Cardiovascular Health Across Sociodemographic Characteristics Submitted to American Journal of Public Health . Joint work with Iheoma U. Iruka , Safoora Masoumi , Rachel M. Vaughn , and Andong Wang . Estimated, using machine learning, how these effects differ across sociodemographic groups. Presentations: American Public Health Association Annual Meeting 2026 , San Antonio, TX (accepted, Nov 2026); Center for AI and Public Health Datathon 2026 , UNC Chapel Hill (talk, 07 Mar 2026) [Slides].	Submitted Aug 2026
Proactive Inpatient Bed Requests for Emergency Department Admissions Submitted to Manufacturing & Service Operations Management . Joint work with Qian Cheng , Nilay Tanik Argon , and Serhan Ziya . Developed data-driven policies for early bed requests using patient-level predictions, reducing boarding time by 30–70% and length of stay by 6–15% in simulations. [arXiv] [SSRN] Presentations: Data Science Day 2026 , School of Data and Information Sciences, UNC Chapel Hill (poster, 24 Sep 2026, second place) [Poster]; YinzOR 2026 Student Conference , Carnegie Mellon INFORMS Student Chapter (poster, 22 Aug 2026, first place) [Poster]; INFORMS Healthcare Conference , Raleigh, NC (talk, 29 Jul 2026) [Slides]; INFORMS Simulation Society Research Workshop (poster, Jul 2026) [Poster]; UNC STOR Graduate Student Seminar (talk, 10 Apr 2026) [Slides]; UNC STOR Department Visit Day (talk, 27 Feb 2026) [Slides]; joint Biostatistics (BIOS)–STOR Student Flashtalks (talk, 20 Feb 2026) [Slides]. The February–April 2026 talks were titled <i>A Decision Framework for Early Inpatient Bed Requests in Emergency Departments</i> .	Submitted Jul 2026
Data Driven Block Replacement Scheduling Submitted to Operations Research . Joint work with Vidyadhar Kulkarni . Developed a data-driven method for learning cost-minimizing preventive maintenance schedules for machine fleets, with provably sublinear regret. [arXiv] [SSRN] [Code] Presentations: INFORMS Annual Meeting , Manufacturing & Service Operations Management (MSOM) Service Operations session honoring Prof. Vidyadhar Kulkarni's retirement, San Francisco, CA (invited talk, Nov 2026); UNC STOR Graduate Student Seminar (talk, 24 Mar 2026) [Slides].	Submitted Jun 2026

Awards & Honors

Finalist , Three Minute Thesis (3MT) 2026 , UNC Graduate School One of 10 finalists selected university-wide, for <i>Admitted, and still waiting...</i> [Announcement]	Oct 2026
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Second Place, Poster Competition, Data Science Day 2026, School of Data and Information Sciences, UNC Chapel Hill For <i>Proactive Inpatient Bed Requests for Emergency Department Admissions</i> , joint work with Qian Cheng , Nilay Tanik Argon , and Serhan Ziya [Poster]	Sep 2026
First Place, Poster Competition, YinzOR 2026 Student Conference, Carnegie Mellon INFORMS Student Chapter For <i>Proactive Inpatient Bed Request Policies for Emergency Departments</i> , joint work with Qian Cheng , Nilay Tanik Argon , and Serhan Ziya [Poster]	Aug 2026
Honorable Mention, Tom Ten Have Poster Competition, American Causal Inference Conference 2026 For the latebreaker poster <i>Estimation of a Common Local Average Treatment Effect with Multiple Instruments</i> , joint work with Patrick Lopatto and P. M. Aronow	May 2026
Best Methodological Rigor, Center for AI and Public Health Datathon 2026, UNC Chapel Hill For <i>Machine Learning-Based Heterogeneous Causal Effects of Social Determinants of Health on Cardiovascular Health Across Sociodemographic Characteristics</i> [Certificate] [Slides]	Mar 2026
Dr. Xin Ge Graduate Student Excellence Award, UNC Chapel Hill Recognizing the best-performing first-year PhD student in the STOR department (see p. 15 of the department newsletter)	Dec 2025
All India Rank 10, Indian Statistical Institute M.Stat. Entrance Examination (nationwide)	May 2022
SHRIRAM Scholarship, Chennai Mathematical Institute Full tuition waiver and monthly stipend for all six semesters	Aug 2019
Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow (SX & SA streams), Dept. of Science & Technology, Govt. of India	2018, 2019
National Talent Search Examination (NTSE) Scholar, National Council of Educational Research and Training	2017
Global 3rd Prize, NASA/National Space Society Space Settlement Contest Presented at the International Space Development Conference , Toronto, Canada	May 2015

Teaching Experience

Instructor of Record, STOR 155: Intro to Data Models & Inference, UNC Chapel Hill Course website Featured in INDY Week for this course	Aug–Dec 2026
Bootcamp Instructor (Real Analysis & Linear Algebra), UNC STOR Lead instructor for incoming PhD students; co-taught with Dr. Panagiotis Andreou . [Course Material]	Aug–Sep 2025
Graduate Teaching Assistant, UNC Chapel Hill STOR 635 (Probability II), Spring 2026, Prof. Sayan Banerjee STOR 634 (Probability I), Fall 2025, Prof. Benjamin Seeger STOR 455 (Methods of Data Analysis), Summer 2025, Kendall Thomas STOR 155 (Intro to Data Models & Inference), Summer 2025, Tianzhu Liu STOR 445 (Stochastic Modelling), Spring 2025, Vidyadhar Kulkarni STOR 435 (Introduction to Probability), Fall 2024, Prof. Mariana Olvera-Cravioto	Aug 2024 – Present
Undergraduate Teaching Assistant, Chennai Mathematical Institute Complex Analysis , Jan–May 2022, Prof. B. V. Rao Analysis III , Sep–Dec 2021, Prof. Parameswaran Sankaran Functional Programming using Haskell , Sep–Dec 2021, Prof. S. P. Suresh Analysis I , Sep–Dec 2021, Prof. Rajeeva Karandikar Analysis Review Sessions , Aug–Sep 2021, Prof. B. V. Rao & Prof. Rajeeva Karandikar Probability Theory , Apr–Jul 2021, Prof. Parameswaran Sankaran Analysis II , Apr–Jul 2021, Prof. Sukhendu Mehrotra Functional Programming Using Haskell , Dec 2020–Mar 2021, Prof. S. P. Suresh Analysis I , Dec 2020–Mar 2021, Prof. Purusottam Rath	Dec 2020 – May 2022
Teaching Assistant, National Programme on Technology Enhanced Learning (NPTEL) Introduction to Haskell Programming (Prof. S. P. Suresh); Functional Analysis (Prof. S. Kesavan, Institute of Mathematical Sciences)	Jul 2021 – Apr 2024

Other Projects

Data Science Intern, [24]7.ai, Bangalore Developed an optimization algorithm for the hospitality sector under Dr. Pratibha Moogi	Jun–Jul 2019
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Reading Project: Random Matrix Theory , Indian Institute of Science, Bangalore Supervised by Prof. Manjunath Krishnapur ; spectral distributions via method of moments and Stieltjes transforms	May–Aug 2021
Reading Project: Dirichlet’s Class Number Formula , Indian Statistical Institute, Bangalore Supervised by Prof. B. Sury ; Dedekind zeta functions and Dirichlet L-functions	May–Jul 2020

Talks, Posters & Outreach

Upcoming

- *Admitted, and still waiting...*, [Three Minute Thesis \(3MT\) Finals](#), [UNC Graduate School](#), Sonja Haynes Stone Center (20 Oct 2026)

Research Talks

- *A Survey of Percolation Theory*, [Chennai Mathematical Institute Mathematics Seminar](#) (20 Mar 2024)

Poster Presentations

- *Estimation of a Common Local Average Treatment Effect with Multiple Instruments*, [American Causal Inference Conference \(ACIC\) 2026](#), Salt Lake City, UT (14 May 2026); latebreaker poster, **Honorable Mention** in the [Tom Ten Have poster competition](#); joint work with [Patrick Lopatto](#) and [P. M. Aronow](#)

Outreach Talks

- *The Mystery of “Why”*, [ROOTS 2025](#), [Raising A Mathematician \(RAM\) Foundation](#) (01 Nov 2025) [\[Abstract\]](#) [\[Slides\]](#)
- *The More You Know, the More You Know You Don’t Know*, [ROOTS 2024](#), [RAM Foundation](#) (09 Nov 2024) [\[Abstract\]](#) [\[Slides\]](#)
- *A Glimpse into Probability*, [ROOTS 2023](#), [RAM Foundation](#) (17 Nov 2023) [\[Abstract\]](#) [\[Slides\]](#)
- *Experiments in Probability*, [RAMTP & Epsilon India 2022](#), [RAM Foundation](#) (19 Feb 2022) [\[Abstract\]](#) [\[Slides\]](#) [\[Video\]](#)
- *The Secure Pattern*, [ROOTS 2021](#), [RAM Foundation](#) (14 Nov 2021) [\[Abstract\]](#) [\[Slides\]](#)
- *From Sports to Courts — The Omnipresence of Math*, [ROOTS 2021 Lecture Series](#), [RAM Foundation](#) (09 Oct 2021) [\[Abstract\]](#) [\[Slides\]](#) [\[Video\]](#)
- *Length at Length*, [Online Math Club](#) (25 Sep 2021) [\[Abstract\]](#) [\[Handout\]](#) [\[Video\]](#)
- *Am I Correct? Probably! Tales of the Use and Misuse of Math*, [DhiMath](#) Public Lecture (09 Sep 2021) [\[Abstract\]](#) [\[Slides\]](#) [\[Video\]](#)

Conferences & Schools Attended

2026 NC Triangle Research Regional Exchange (NC Analytics and AI Day), NC State, Raleigh, NC One-day conference on business analytics and AI, organized by the North Carolina INFORMS chapter and NC State Industrial and Systems Engineering	Sep 2026
Conference on Extreme Value Analysis 2025 , UNC Chapel Hill Volunteered at this international conference; an introduction to extreme value theory and its applications across climate science, public health, and finance	Jun 2025
DAIR³: Data and AI Intensive Research with Rigor and Reproducibility , Jackson State University National Institutes of Health (NIH)-supported weeklong bootcamp on rigorous and reproducible biomedical data science: ethical AI, data management, analytical design, and reproducible workflows	May 2025
Mini-course by Prof. Remco van der Hofstad , UNC Chapel Hill Short course on <i>Dynamic Processes on Networks</i>	Oct 2024
Advanced Instructional School in Stochastic Processes (AISSP) , NISER Bhubaneswar Three-week school on Brownian motion, martingales, and measure theory, organized by the National Centre for Mathematics [Notes & Problem Sets]	Jun 2023

Service & Leadership

STOR Liaison to the Department of Biostatistics , UNC STOR Facilitate communication and plan activities between the STOR and BIOS departments	Sep 2026 – Present
Lead Organizer, STOR Graduate Student Showcase , UNC Research Week Organizing the department’s afternoon of ten PhD research posters and talks	Oct 2026
Peer Reviewer, <i>Health Care Management Science</i> , <i>Proceedings of the American Mathematical Society</i> , and <i>Involve, a Journal of Mathematics</i>	
Mentor, Twoples – mentoring an advanced undergraduate in High Dimensional Probability	Oct 2023
Event Coordinator & Head Organizer, STEMS & Tessellate , Chennai Mathematical Institute End-to-end planning of Chennai Mathematical Institute’s annual inter-collegiate math & science festival; organized STEMS lecture series & camps featuring leading researchers from top institutions worldwide	Sep 2019 – Mar 2022
Student Treasurer, Chennai Mathematical Institute	Mar 2020 – May 2022

Volunteering

UNC Science Expo , Morehead Planetarium	Apr 2025, Apr 2026
Represented STOR department with interactive exhibits on a mind reader, the Monty Hall problem, Galton board, and the Traveling Salesman Problem	
UNC Engineers Week – represented STOR dept. with operations research and healthcare operations demos	Feb 2026
Teaching Volunteer, Focus A1 Academy for the National Institute of Open Schooling (NIOS)	May–Jun 2022
Taught Descriptive Statistics (Grades 10 & 12) and Probability (undergraduate), 4 hrs/week	
Math Olympiad Mentor, Rural Odisha (30 students)	May–Jun 2021
Mentored students online for the Regional Mathematical Olympiad on Inequalities and Geometry	
Student Volunteer, Bridge India Foundation	2021
Coordinated visits to Samarathanam Trust for the Disabled and Sri Rakum School for the Blind; helped build a handheld device for visually challenged individuals to identify currency denominations	

Selected Coursework

UNC Chapel Hill: Causal Machine Learning, Precision Medicine, Reinforcement Learning & Markov Decision Processes, Simulation Modeling

Indian Statistical Institute: Martingales, Percolation Theory, Brownian Motion, Functional Analysis, Time Series Analysis, Large Sample Theory

Chennai Mathematical Institute: Measure Theory, Real & Complex Analysis, Topology, Algorithms, Theory of Computation, Linear Algebra

Skills

Programming: Python (pandas, NumPy, scikit-learn, PyTorch/TensorFlow), R (tidyverse, ggplot2), SQL

Tools: Git and version control, Excel, $\text{\LaTeX}$

Languages: English (Fluent), Tamil (Native), Hindi, Kannada, Bangla